

JEE MAIN REVISION MASTER: QUADRATIC EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Roots, Coefficients & Nature of Roots (Discriminant)	3 PYQs	Quadratic Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Roots, Coefficients & Nature of Roots (Discriminant)	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$, $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Roots, Coefficients & Nature of Roots (Discriminant) (Moderate Yield — 3 PYQs)

Core Principle: Quadratic equation $ax^2 + bx + c = 0$. Roots $\alpha, \beta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Sum $\alpha + \beta = -b/a$, Product $\alpha\beta = c/a$. Discriminant $\Delta = b^2 - 4ac$: $\Delta > 0$ (real & distinct), $\Delta = 0$ (real & equal), $\Delta < 0$ (complex conjugate roots). Condition for a common root between two quadratics.

■ **Caution / JEE Pitfall:** If coefficients a, b, c are rational and Δ is not a perfect square, roots occur in conjugate surd pairs ($p + \sqrt{q}$ and $p - \sqrt{q}$).

■ **Speed Trick:** Difference of roots: $|\alpha - \beta| = \frac{\sqrt{\Delta}}{|a|}$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "Given that a, b and x are real numbers and $a < b, x < 0$, then..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If α and β are roots of the equation $x^2 = x + 1 = 0$, then $\alpha^2 + \beta^2$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is

- (A) $x^2 - 10x - 16 = 0$
- (B) $x^2 + 10x + 16 = 0$
- (C) $x^2 + 10x - 16 = 0$
- (D) $x^2 - 10x + 16 = 0$

Correct Answer: None

Solution Analysis: Let a and b be the roots of the quadratic equation. Then, the quadratic equation is $x^2 - (a+b)x + ab = 0$ (i)
It is given that $AM=5$ and $GM=4$ i.e. $(a+b)/2=5$ and $\sqrt{ab}=4 \Rightarrow a+b=10$ and $ab=16$. On substituting these values in Eq. (i), we obtain. Hence, $x^2 - 10x + 16 = 0$ is the required equation.